Security Audit Report

Authorized password-strength assessment & storage-security analysis

AUTHORIZED USE ONLY. This report was produced by an authorized password security assessment against SYNTHETIC test data. It must not be used to access accounts without explicit permission. No real user secrets are collected or stored.

Audit Overview

Dataset	strong
Attack type	all
Started	2026-11-02T13:20:44+00:00
Finished	2026-11-02T13:20:44+00:00
Duration (s)	0.230
Accounts audited	20
Algorithms	sha256

Results Summary

Passwords recovered: 0 / 20 (0.0%)

Overall risk: Low Storage grade: Poor

Total candidate attempts: 20

Risk breakdown - Critical: 0 High: 0 Medium: 0 Low: 20

Password Storage Assessment

- Passwords are stored as fast, unsalted hashes -- vulnerable to dictionary, hybrid and rainbow-table attacks. Migrate to a salted, slow KDF (PBKDF2, scrypt, bcrypt or Argon2).

Per-Account Findings

Username	Algo	Cracked	Method	Risk	Recovered
lab001	sha256	No	-	Low	-
lab002	sha256	No	-	Low	-
lab003	sha256	No	-	Low	-
lab004	sha256	No	-	Low	-
lab005	sha256	No	-	Low	-
lab006	sha256	No	-	Low	-
lab007	sha256	No	-	Low	-
lab008	sha256	No	-	Low	-
lab009	sha256	No	-	Low	-
lab010	sha256	No	-	Low	-
lab011	sha256	No	-	Low	-
lab012	sha256	No	-	Low	-
lab013	sha256	No	-	Low	-
lab014	sha256	No	-	Low	-
lab015	sha256	No	-	Low	-
lab016	sha256	No	-	Low	-
lab017	sha256	No	-	Low	-
lab018	sha256	No	-	Low	-

lab019	sha256	No	-	Low	-
lab020	sha256	No	-	Low	-

Recommendations

- Re-hash all stored passwords with a salted, slow KDF (PBKDF2 >= 200k iters, scrypt, bcrypt, or Argon2id).
- Enforce a minimum length of 12+ characters and screen new passwords against a common-password blocklist.
- Enable multi-factor authentication to reduce the impact of any single compromised password.

Method Limitations

- Dictionary: only finds passwords present in the wordlist.
- Hybrid: only rule-expressible mutations; candidate space grows fast.
- Rainbow/lookup: storage grows with coverage and is defeated by salting; this tool uses a direct hash-lookup table, not reduction-chain rainbow tables.
- Results are bounded by the chosen wordlist and single-machine Python speed; a dedicated GPU cracker would be faster.